

Books to Scrape: Book Price & Rating

Analysis – Project Summary

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- Batch :- February 2026
- Course :- Data Science
- Video Link:-

<https://www.loom.com/share/c99c83692d854c82b8644c26411baa54>

Project Overview

This project is about collecting and analysing book data from the **Books to Scrape** website. I collected **1,000 book records** from 50 pages using Python, Requests and BeautifulSoup. The main fields collected were book title, price and rating.

The main aim was to understand the relationship between **book price, rating and title length**, and then present the findings through an interactive Power BI dashboard.

Objective

- collect book data from the Books to Scrape website.
- Clean and prepare the collected data.
- Analyse book price, rating and title length.
- Compare books across different price categories and ratings.
- Create a Power BI dashboard to show the main findings..
- Suggest better business decisions.

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Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- BeautifulSoup
- Google Colab
- Power BI

Project Workflow

- Web Scraping
- Data Collection
- Data Cleaning
- Data Analysis
- EDA

- Price Category Analysis
- Rating Analysis
- Correlation Analysis
- Power BI Dashboard

1. Web Scraping

I used **Books to Scrape** to collect book data using Python, Requests and BeautifulSoup. Data was scraped from **50 pages**, giving **1,000 book records** containing book title, price and rating. The collected data was stored in a Pandas DataFrame and used for further analysis and visualization.

2. Data Preparation

- **1,000 book records** were collected.
- The original data had **3 columns**: title, price and rating.
- Price was converted into numeric format.
- Rating values were converted from words into numbers from **1 to 5**.
- No missing values were found.
- No duplicate records were found.
- Two new features were created:
- **title_length**
- **price_category** – Low, Medium and High.
- Final dataset had **1,000 rows and 5 columns**.

Chart 1 :- Average Book Price by Rating



Actual Output:

- Average book price: **£35.07**
- Rating 4 books have the highest average price: **about £36.1**
- Rating 5 books have an average price of **about £35.4**
- The average prices across all five ratings are quite close.

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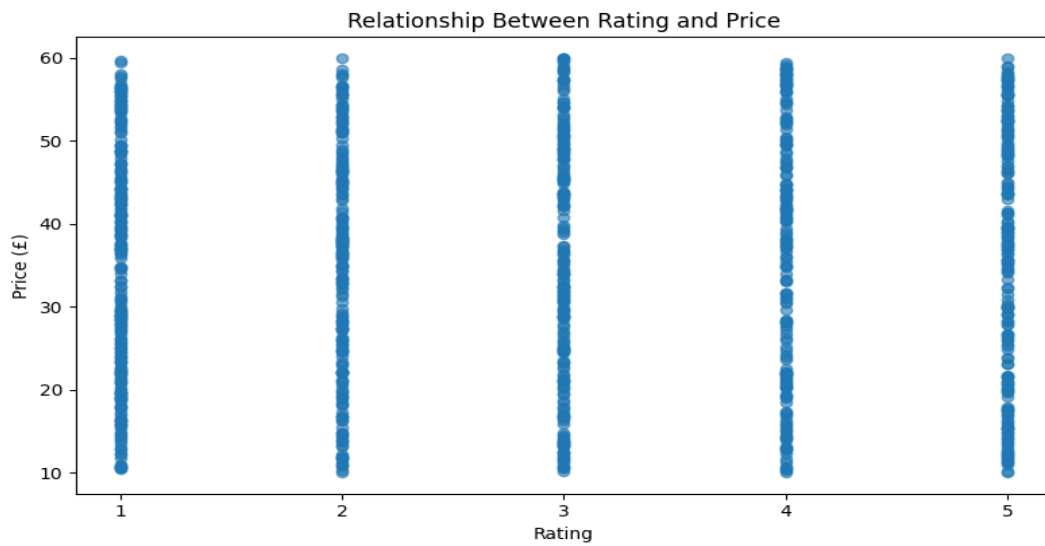
Insight:

The price difference between ratings is very small. Interestingly, **5-star books are not the most expensive**; 4-star books have the highest average price. This shows that a higher rating does not automatically mean a higher price.

Recommendation:

Do not decide book pricing based only on rating. **Price and rating should be checked together** before making a pricing decision.

Chart 2 :- A Rating vs Price Scatter Plot



Actual Output:

- Price ranges from **10.00 to £59.99**
- Correlation between price and rating is only **about 0.03**.

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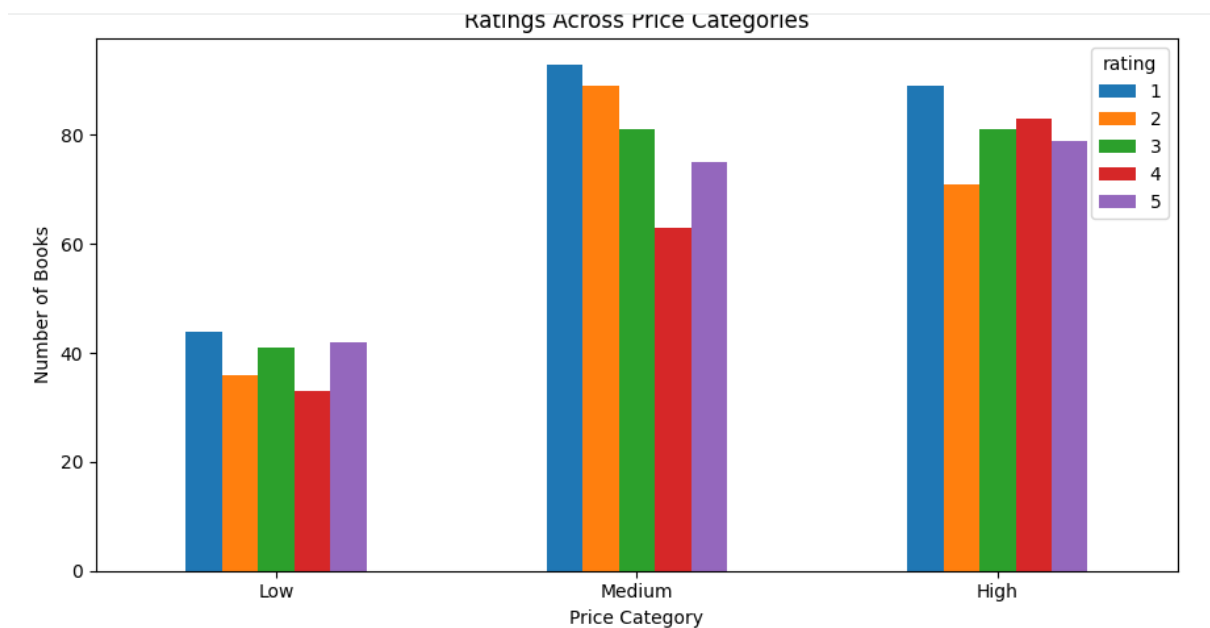
Insight:

The scatter plot does not show a clear upward pattern between price and rating. Books with higher prices are not consistently getting higher ratings. So, **price is not a strong indicator of rating in this dataset**.

Recommendation:

Avoid assuming that expensive books are better rated. For book comparison, **look at rating and price separately** instead of using price as a quality indicator.

Chart 3.-Rating Distribution Across Price Categories



Actual Output:

- The dataset contains **1,000 books**.
- **High and Medium** price categories contain around **400 books each**.
- **Low** price category contains around **195 books**.
- 1-star rating is the most common with **226 books**.
- 4-star rating is the least common with **179 books**.

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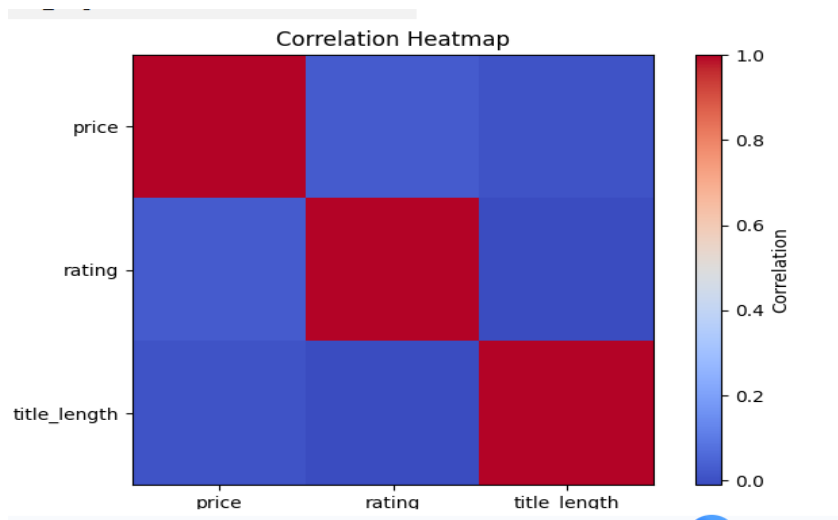
Insight:

The catalog has noticeably fewer books in the **Low-price category** compared with Medium and High. At the same time, **1-star books form the largest rating group**. So, the catalog has more books in the middle and higher price ranges, while low ratings are also quite common.

Recommendation:

If the aim is to provide more budget options, the **Low-price category can be reviewed and expanded**. Also, the large number of 1-star books should be looked at separately to understand the rating mix.

Chart 4. Correlation Heatmap



Actual Output:

- Price vs Rating correlation: **about 0.03**
- Price vs Title Length correlation: **about 0.01**
- Average title length: **39.12 characters**.

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Practical Insight:

Both relationships are very weak. This means **rating and title length do not explain much about book price** in this dataset. Title length alone is not useful for understanding why one book costs more than another.

Recommendation:

Do not use title length or rating alone for pricing decisions. For deeper analysis, additional fields such as **sales, number of reviews, or stock information** would be useful because the current data cannot show actual sales or revenue performance.

Power BI Dashboard

Page 1 – Book Overview



Actual Output:

The dataset has **1,000 books**, with an average price of **35.07**, average rating of **2.92 stars**, and average title length of **39.12 characters**.

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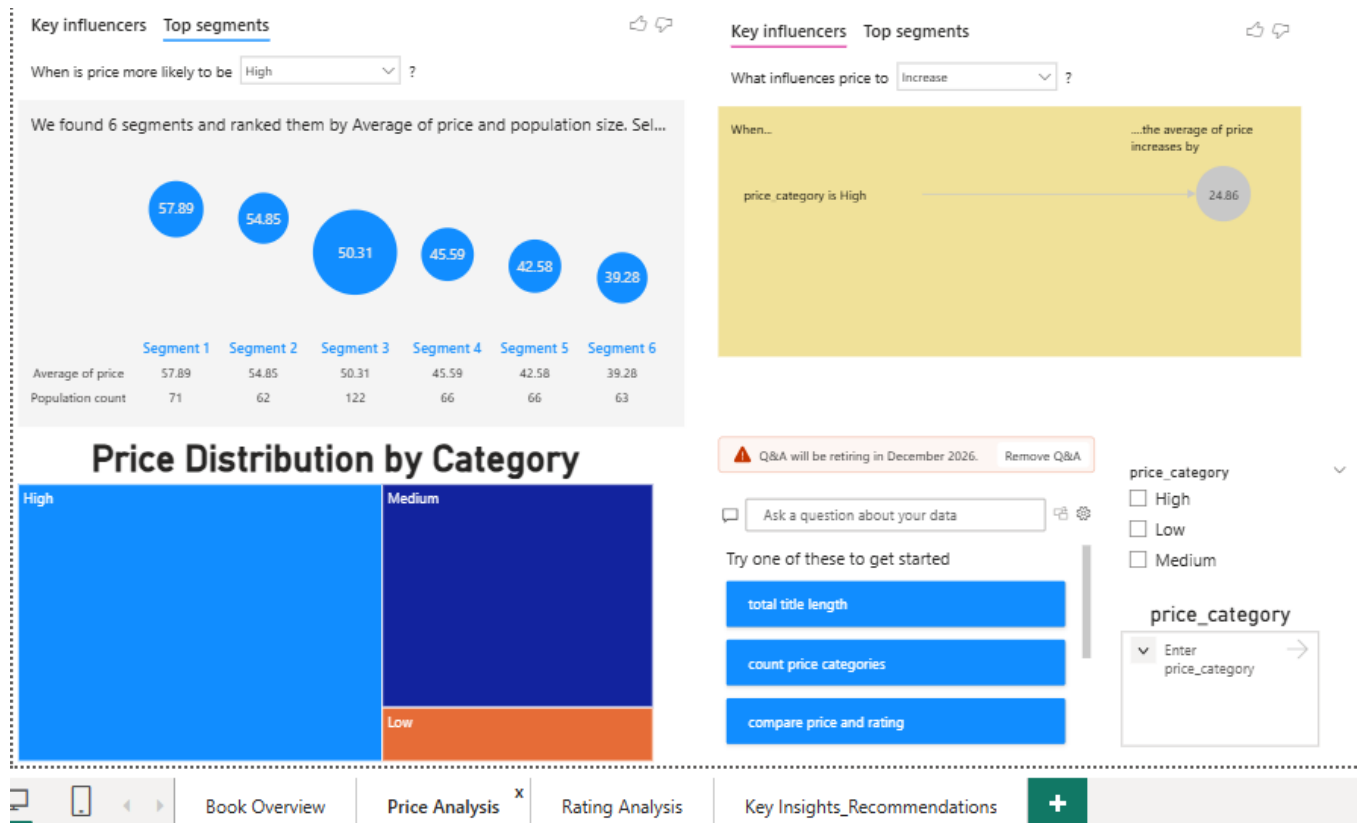
Insight:

This page gives a quick overall view of the book catalog. The average rating is below 3 stars, while prices cover a wide range from 10 to £59.99.

Recommendation:

Use this page as the **starting point** before checking detailed price or rating patterns.

Page 2 – Price Analysis



Actual Output:

Books are divided into **Low, Medium and High** price categories. Medium and High contain around **400 books each**,

while Low contains around **195 books**.

Practical Insight:

Most books are in the Medium and High price groups. The Low-price group is much smaller, so budget choices are less represented in the catalog.

Recommendation:

Keep the price-category comparison visible and **review the Low-price range** if the goal is to provide more affordable choices.

Page 3 – Rating Analysis

Key Insights

- **Price Peak at 4-Star Rating:** Books with a 4-star rating have the highest average price (\$36.1), whereas 5-star books have a slightly lower average price (\$35.4).
- **Dominant Category:** The "High" price category represents the largest share of books in the catalog.
- **Low-Price Gap:** The "Low" price segment has the smallest proportion of titles, indicating fewer budget options.

Recommendations

- **Review Pricing for Top-Rated Books:** Re-evaluate the pricing strategy for 5-star titles to align them closer to the higher average price observed in 4-star books.
- **Expand Budget Offerings:** Add more lower-priced titles to balance the catalog and cater to price-sensitive readers.
- **Focus Catalog Strategy on High Tier:** Maintain strong stock in the high-price tier, as it forms the primary foundation of the current inventory.

Book Overview

Price Analysis

Rating Analysis

Key Insights_Recommendations x +

Actual Output:

- Average rating: **2.92 stars**
- 1-star books: **226**
- 4-star books: **179**
- Average price of 4-star books: **about 36.1**
- Average price of 5-star books: **about 35.4.**

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Practical Insight:

The rating distribution is not balanced, with 1-star books being the largest group. Also, 5-star books are slightly cheaper on average than 4-star books, which again shows that **higher rating does not automatically mean higher price.**

Recommendation:

Use the rating filter together with price when comparing books. **Do not treat rating as a direct pricing factor.**

Overall takeaway

The main point from the project is that **book price and rating have almost no relationship (correlation ≈ 0.03)**. The catalog has a wide price range, fewer Low-price books, and a large number of 1-star books. So the useful business action is not simply to increase prices for highly rated books; instead, **price categories and rating patterns should be checked together.**

Thank You